2 RELATED WORK

Our work is inspired by the intersection of several key areas: social influence in LLMs, the misinformation and factual robustness in LLMs, and LLM-based multi-agent systems.

Social influence in LLMs. LLMs exhibit susceptibility to social influence, displaying conformity (Weng et al., 2025), debate (Du et al., 2023), and sycophancy (Perez et al., 2023; Li et al., 2025a) tendencies. Other work has investigated how LLMs can be persuaded by user arguments (Xu et al., 2024). However, existing work focuses on short-term, in-context compliance. We advance the field by introducing *long-term memory solidification*—measuring whether socially-induced false beliefs become internalized into stable memories, a defining characteristic of the Mandela effect.

LLM-based multi-agent systems. LLM-based multi-agent systems (Wu et al., 2024; Hong et al., 2024; Chen et al., 2024a; Wu et al., 2025a; Ye et al., 2025) utilize multiple LLM agents (Wang et al., 2024) to combine their collective intelligence and specialized skills, enabling robust and scalable solutions for complex tasks (Guo et al., 2024). Agents typically engage in iterative discussions and collaborative decision-making, mirroring the dynamics of human teams (Park et al., 2023; Zhang et al., 2024). Despite these advancements, exploration of collective cognitive biases remains limited.

Misinformation and factual robustness in LLMs. Most research focuses on the factuality of individual LLMs. A primary focus is on mitigating hallucinations, where models generate non-factual information (Elazar et al., 2021; Ji et al., 2023; Chen et al., 2024b; Ju et al., 2024; Liu et al., 2025b; Li et al., 2023b; 2025b). Some involve external verification of the outputs with trusted knowledge bases for fact-checking (Tang et al., 2024). In LLM-based multi-agent systems, tse Huang et al. (2025) investigates the robustness of systems against malicious agents that intentionally inject incorrect information to disrupt group consensus. However, these studies mainly treat factual errors as technical failures. Our work addresses *socially-induced misinformation*, where false memories are caused by persuasive social contexts.